

CENTA Carbon Fibre Shafts

Type P for High Torque / Baureihe P für hohe Drehmomente

HIGH-PERFORMANCE LIGHTWEIGHT SHAFT SYSTEMS

CENTA carbon fibre composite shafts offer significant weight savings of up to 70% compared to conventional steel shafts. Specifically engineered for industrial applications demanding extreme torque capacities and high rotational speeds, Type P shafts minimize mass moments of inertia while providing outstanding torsional stiffness.

KEY ADVANTAGES

- **Laminate with High Speed Suitability:** Optimized fiber orientation allows exceptional speed boundaries and minimal whirling risk over long spans.
- **Substantial Load Reductions:** Lightweight construction minimizes weight on intermediate bearings and foundation supports.
- **Corrosion Resistant:** Built with carbon-glass composites perfect for harsh marine, cooling tower, and industrial environments.
- **Customized Spans:** Allows massive distance-between-shaft designs without the need for multiple intermediate support bearings.

TECHNICAL DRAWING & CONSTRUCTION LAYOUT

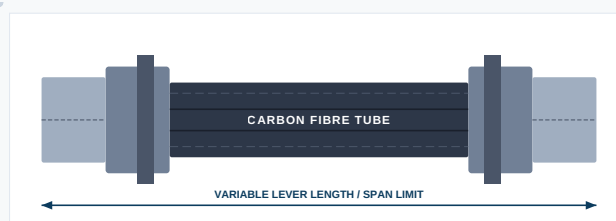


Figure 1: Typical schematic assembly of CENTA Type P carbon fibre high-torque intermediate shaft, exhibiting lightweight mid-section tube integrated with heavy-duty end fittings.

TECHNICAL SPECIFICATIONS TABLE

Complete catalog performance ratings and dimensions for Baureihe P (Series P).

Shaft Size	Nominal Torque [kNm]	O.D. [mm]	Mass per meter length [kg]	Mass moment of inertia / m length [kgm ²]	Torsional stiffness / m length [MNm/rad]	Mass per end fitting* [kg]	Intermediate shaft bearing Ø [mm]	Nominal Ø of bulkhead seal [mm]
Laminate with high speed suitability								
CFRP5P162	5	162	3,7	0,023	0,083	5,6	75	200
CFRP6P163	6	163	4,1	0,026	0,092	5,6	80	200
CFRP8P164	8	164	4,5	0,028	0,101	10	85	200
CFRP10P166	10	166	5,3	0,034	0,120	10	95	200
CFRP12P214	12	214	5,9	0,064	0,229	8,7	100	230
CFRP15P216	15	216	6,9	0,076	0,271	8,7	105	260
CFRP20P268	20	268	8,7	0,148	0,527	14	120	290
CFRP25P269	25	269	9,3	0,159	0,568	14	125	320
CFRP30P270	30	270	9,9	0,171	0,609	23	135	320
CFRP40P317	40	317	12,4	0,296	1,057	34	150	350
CFRP50P318	50	318	13,2	0,315	1,124	34	160	350
CFRP60P321	60	321	15,5	0,373	1,331	41	170	350
CFRP80P371	80	371	18,0	0,584	2,083	71	190	410
CFRP100P375	100	375	21,5	0,707	2,522	71	200	410
CFRP120P424	120	424	23,5	0,996	3,553	106	210	470
CFRP150P430	150	430	29,5	1,273	4,541	106	230	470
CFRP200P525	200	525	30,4	1,997	7,122	216	250	560
CFRP250P532	250	532	39,2	2,609	9,308	216	270	590
CFRP300P537	300	537	45,5	3,062	10,92	255	290	590
CFRP400P612	400	612	53,4	4,704	16,78	306	320	680
CFRP500P705	500	705	57,1	6,748	24,07	490	340	~750
CFRP600P712	600	712	68,8	8,225	29,34	490	360	~750
CFRP800P725	800	725	91,0	11,08	39,54	490	400	~800

** Mass per end fitting values relate to standard manufacturing configurations. Continuous updates and customizable alterations may adapt these indices over time.*

Procurement & Technical Support in India:

For standard requests, precise engineering support, torque alignment calculation, or official quotations, please get in touch with our authorized engineering partner network:

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